Skony — Track 2A OpenParlData: Technical Report

Repository: <https://github.com/ahmarkhhan/sfedly> (built from the HackApertus project template). Build and completion evidence: BUILD_STATUS. Scoped affair contracts and offline evaluation tooling are implemented; human-reviewed cross-body evaluation remains pending. Response caching, batch checkpoints, saved-result import and gold-export safeguards are implemented. Docker packaging and a synthetic practice replay exist; actual container execution is still unverified. Pipeline evidence.

Status: extraction and local review implementation, 16 October 2026. One real-document pilot is complete; dataset-level extraction accuracy has not been measured. Measured run evidence: RESULTS.

Judging criteria and where they are addressed

Criterion (official brief wording)	Where it is addressed	State
Insight: a well-founded conclusion on common-schema feasibility; partial or negative counts	Honest limits; RESULTS	Partial: a common raw-field representation proved feasible for the eight-field slice tested; the unresolved seam is question/answer pairing and role assignment across bodies
Breadth: demonstration across document types, languages and levels of government	RESULTS §Measured runs; corpus integrity table in PIPELINE_VERIFICATION	Narrow but real: DE + FR, two bodies (municipal Nidau, cantonal Genève), born-digital plus one scanned/OCR-layer document; a 44-PDF hash-verified corpus awaits review
Traceability and faithfulness: no invented values; span-level provenance; flag rather than guess	Implemented first slice; `src/skony/verify.py` (abstains on unlocated, fabricated or ambiguous evidence); strict-validation rejection tests in `tests/test_evidence.py`	Implemented: every non-null field carries document ID, page, original quote and character offsets; unverifiable values abstain
Built with Apertus	`src/prompts/extract.txt`; RESULTS (recorded `swiss-ai/Apertus-v1.5-70B` runs with token and latency observations)	Yes, at CSCS; development-time only, not redistributed
Reusable outputs: public repository with converter, prompts, validator and README samples	README quick-start; RESULTS §Reproduce; hash-checked synthetic fixture in `data/sample/`; current offline engineering suite	Public synthetic replay runs without credentials; the real DE/FR cache replay requires privately retained inputs
Judge execution: `make run` in Docker	root and track `Makefile`, `Dockerfile`, `compose.yaml`, `scripts/check_docker.py`, CI workflow `.github/workflows/docker-verify.yml`	Packaged, unverified on the development host (no Docker/WSL there); the clean public-source start passes outside a container

Problem and contribution

Investigate which parliamentary affair fields can share a common representation across bodies and languages. Preserve original wording, explicitly abstain when evidence cannot be located, and record where the shared schema cannot represent the source faithfully. Our partial conclusion: a common raw-field representation is feasible for the eight-field slice tested — fields were located with original wording and spans across one DE municipal and one FR cantonal document — while question/answer pairing and role assignment across bodies remain the unresolved seam.

Implemented first slice

PDF → page text and digest → Apertus schema-constrained prompt → strict JSON validation → evidence lookup → review-required JSON. The output has eight raw fields. A draft controlled-vocabulary layer maps labels to 33 stable term IDs (229 DE/FR/EN aliases) while preserving raw values and explicit unmapped outcomes. A structural inventory detects annex markers, images and table regions with provenance; it does not interpret their values. The research explorer searches imported PDF text and exports cited briefs. Scoped v0.2 review, comparison, migration, gold eligibility and offline evaluation tooling are implemented; human gold and measured semantic accuracy remain pending.

Each non-null result includes a document ID, page number, original quote and character offsets into the unchanged extracted page text. `source_located` means a string was found; it does not establish correct field assignment or faithfulness of question/answer boundaries. All such results require review.

Validation so far

245 engineering tests and 16 interface-controller tests passed on local Python 3.11 / Node 24 (RESULTS). The client was exercised with mocked HTTP transports and the real CSCS service. Tests cover fabricated quotes/values, wrong page attribution, repeated quotes, missing evidence, textless/oversize input, malformed schemas, truncated/mismatched endpoint responses, line-break span recovery and obviously corrupt submitter OCR. Review tests also cover persistence, stale revisions, fabricated correction rejection, explicit absence, visual transcription, audit snapshots, model reruns preserving human decisions, reviewed-only export and local route/secret isolation.

New execution checks cover replay blocking live calls, synthetic-fixture hashes, restart persistence and mapped localhost Host/Origin checks. A clean public-source copy starts without private data/keys and serves the PDF/image and reviewed export. The bundled practice data are authored examples, not model output or evaluation gold.

The batch/cache checks cover full effective-request identity, corrupt-cache rejection, schema-invalid response replay, actual call accounting, preserved failed downloads, unchanged manifests, interrupted resume and idempotent workbench import. Gold-export checks exclude synthetic/QA records and prevent stale labels from a previous export directory entering a new manifest.

An approved real-data engineering proof processed one DE Nidau PDF and one FR Genève PDF using Apertus 1.5 70B at CSCS. The first run made two calls, reporting 3,953 total tokens and 13.594/9.406 seconds. Offline replay, with no credential and an HTTP-rejecting transport, used two cached responses and produced identical verified results with zero model calls. One failed-fetch row remained in both ledgers. This demonstrates pipeline reuse, not semantic accuracy or multilingual benchmark coverage. Both results were imported without altering existing review decisions. The store has 12 decisions across four documents, one mechanically complete; it is not yet a frozen, competence-checked evaluation gold set.

On one scanned Nidau postulate with an existing OCR layer, the 8B pilot first copied the schema and then returned inconsistent null/evidence combinations. Both were rejected. The 70B pilot returned valid fields. Replaying that recorded response with verifier `evidence-v0.2` located the title, type, printed date and request; it abstained on the garbled submitter and absent answer/decision fields. The 70B call took 7.234 seconds and used 1,725 tokens. These are single-call observations, not a benchmark or a gold-set result.

Planned evaluation

Use separate development and held-out human-reviewed documents spanning German and French and multiple parliamentary bodies. Compare a fair simple baseline, raw Apertus extraction and extraction with evidence checks. Report field precision/recall, abstention/coverage, source-location accuracy, question/answer pairing, latency and actual token usage. Scans and minority layouts must remain visible in the report rather than being filtered out of the denominator.

Honest limits

Each bullet states a limitation, its consequence and where the evidence is. This section, not the feature list, is the insight contribution.

- **Two documents are not coverage.** The DE/FR batch contains two team-selected

real PDFs; the earlier scanned Nidau pilot is recorded separately above. Nothing measures accuracy on unseen documents; breadth above is corpus breadth, not measured extraction breadth. Evidence: RESULTS.

- **In-sample tuning.** The prompt (`raw-v0.2`) and verifier (`evidence-v0.2`)

were shaped on the same pilot documents that demonstrate them. The replay results are self-consistent by construction, not evidence of generalisation. Evidence: PIPELINE_VERIFICATION.

- **No gold, no rates.** Twelve human field decisions exist but are not

competence-checked, and one reviewed source language is unrecorded. Precision, recall and abstention rates would be fabricated today, so none are presented. Evidence: RESULTS §Review and gold-set state.

- **Located is not correct.** `source_located` proves a string match, not

semantic assignment: a value can occupy the right span and the wrong field. Every located field therefore requires review; question/answer pairing is unvalidated. Evidence: Implemented first slice.

- **Docker is packaged, not proven.** No container has been built on the

development host (no Docker/WSL); the CI workflow and `scripts/check_docker.py` are ready but unexecuted there. The official judge path could therefore fail on first contact; `scripts/check_clean_start.py` covers the non-container path. Evidence: PIPELINE_VERIFICATION.

- **One previously flaky test.** `test_mutation_requires_token_and_matching_origin`

in `tests/test_review.py` failed intermittently on the development host; it passed five consecutive runs on 16 October and the fresh-clone suite is passing at the time of that earlier check. Named so a judge re-running it is not surprised. Evidence: `python -m unittest tests.test_review.ServerTests.test_mutation_requires_token_and_matching_origin`.

- **Heuristic roles and languages.** Corpus role and level tags are heuristic

hints; 13 of 45 manifest rows are unknown-language. Level-of-government breadth claims wait for reviewed metadata. Evidence: BUILD_STATUS.

- **Conservative matching has blind spots.** Whitespace-only recovery cannot

bridge OCR reflow, and the corrupt-name marker does not catch all corruption. Some real values will read as abstentions until review — accepted as the price of never guessing. Evidence: `src/skony/verify.py`.

The local workbench displays PDF page images, source text and highlighted spans. Reviewers can confirm, correct, record absence or undo. Decisions are separate from model suggestions in SQLite, with before/after audit history. Visual transcriptions are explicitly labelled and carry no claimed text offsets. Exports contain reviewed fields only, plus a list of remaining unreviewed fields. Reviewer identity is a self-entered label; multi-user authentication is not implemented.

OCR and automatic multi-page field assembly remain unimplemented. Controlled-vocabulary mapping, structural inventory and a source-text research explorer now exist; their synthetic engineering checks do not establish cross-body semantic comparability. Gold-set evaluation still requires independent human annotation. The Docker runner is packaged but has not been built/run in this environment. Whitespace-only matching recovers some layout differences but preserves wording, punctuation and original offsets. Other OCR/layout differences still require review; the OCR marker heuristic does not detect all corrupt names. Copied text can still be assigned to the wrong semantic field. Model identity must be confirmed against the provider's documented served-model mapping.

Deployment and licensing

The application uses a configurable Apertus endpoint. A Swiss endpoint and self-hosted endpoint are deployment targets; air-gapped operation has not been tested.

Component	Licence	Note
Application code (this repository)	Apache-2.0	per event terms
Documentation and this report	CC-BY-4.0	per event terms
Bundled synthetic fixture (`data/sample/`)	CDLA-Permissive-2.0	authored; fictional people and parliament
Runtime: pydantic, httpx, pypdf, PyMuPDF (pinned in `requirements.txt`)	MIT / BSD-3-Clause; PyMuPDF AGPL-3.0 or commercial	PyMuPDF licensing must be satisfied when distributing the application; see upstream licence
Apertus v1.5 via the CSCS endpoint (development time)	Apache-2.0 (model)	not redistributed
OpenParlData source PDFs (private corpus)	source-specific terms	not redistributed; a source manifest must state item-level licences before any redistribution or before a dataset licence is applied to our own annotations
Container base (`python:3.11-slim`, Poppler)	Debian package terms	runner image only